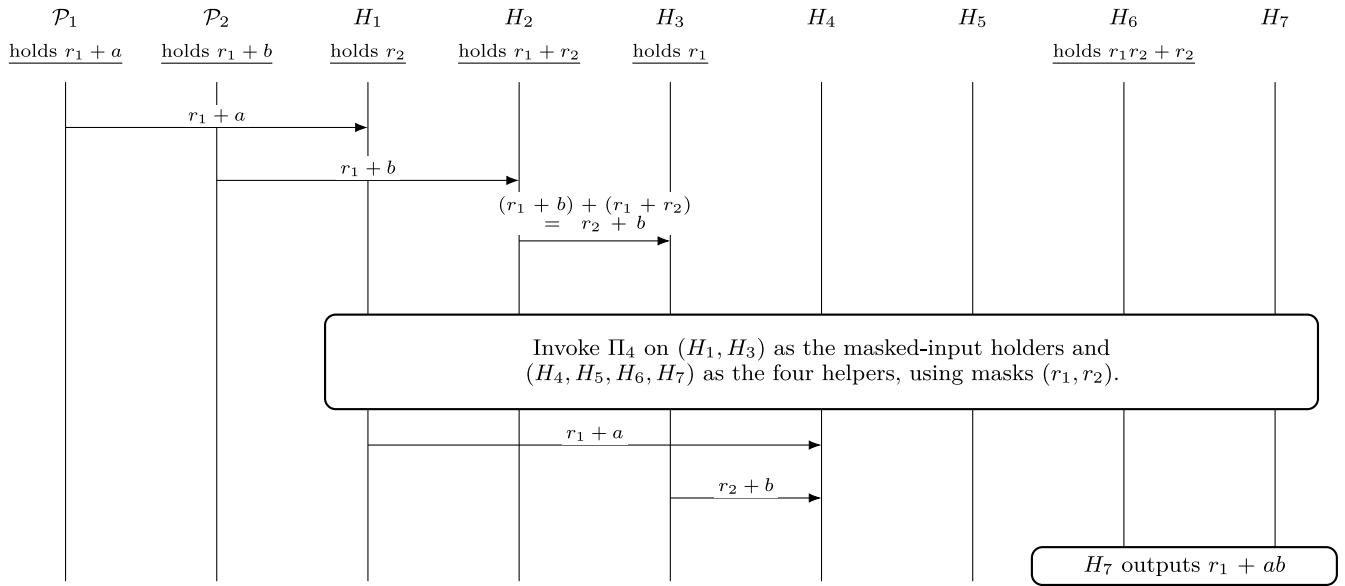
$\mathcal{P}_1$
holds $r_1 + a$
$\mathcal{P}_2$
holds $r_1 + b$
H_1
holds r_2
H_2
holds $r_1 + r_2$
H_3
holds r_1
H_4
H_5
H_6
holds $r_1 r_2 + r_2$
H_7
$r_1 + a$
$r_1 + b$
$(r_1 + b) + (r_1 + r_2)$
$= r_2 + b$
Invoke Π_4 on (H_1, H_3) as the masked-input holders and
(H_4, H_5, H_6, H_7) as the four helpers, using masks (r_1, r_2).
$r_1 + a$
$r_2 + b$
H_7 outputs $r_1 + ab$